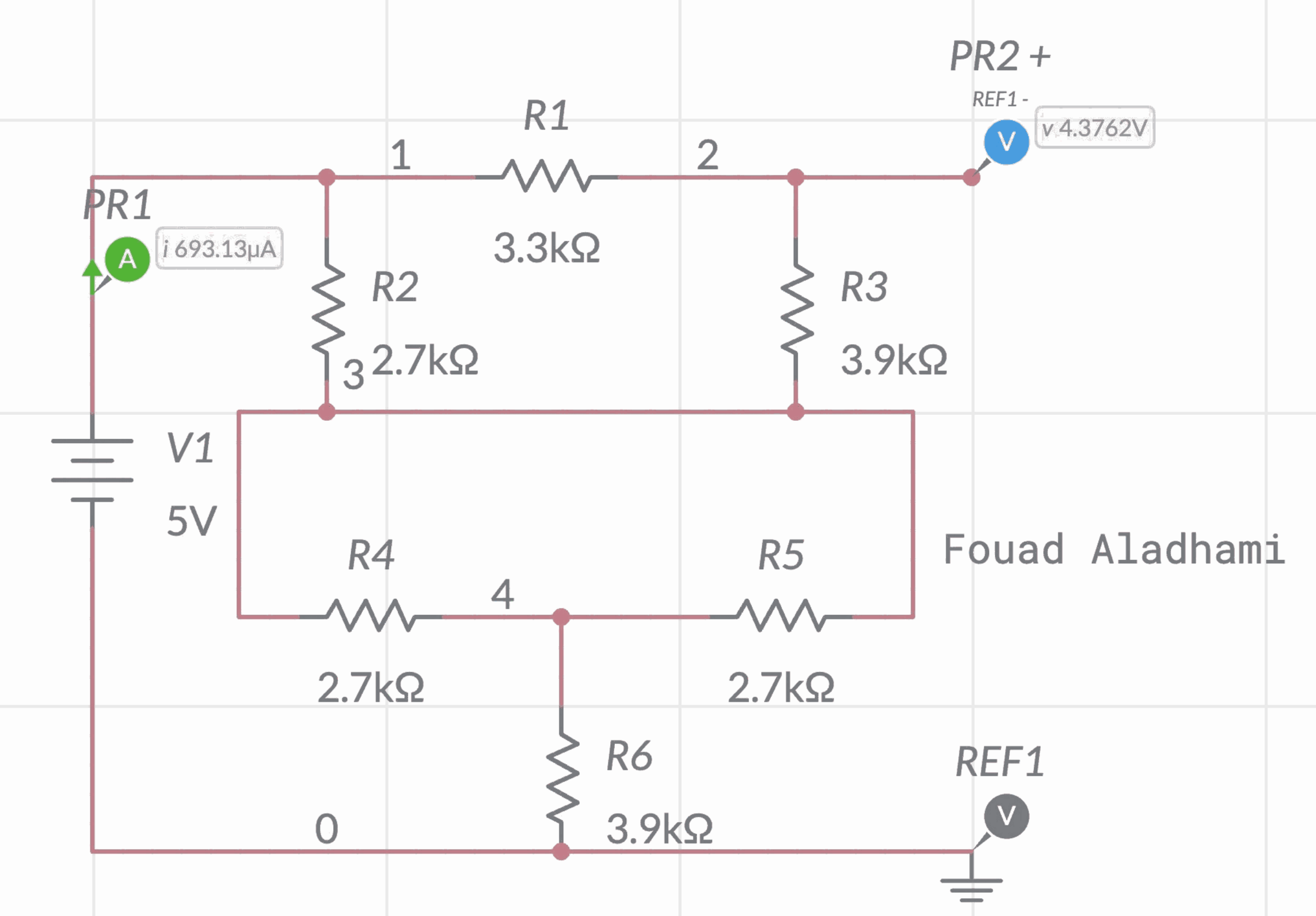
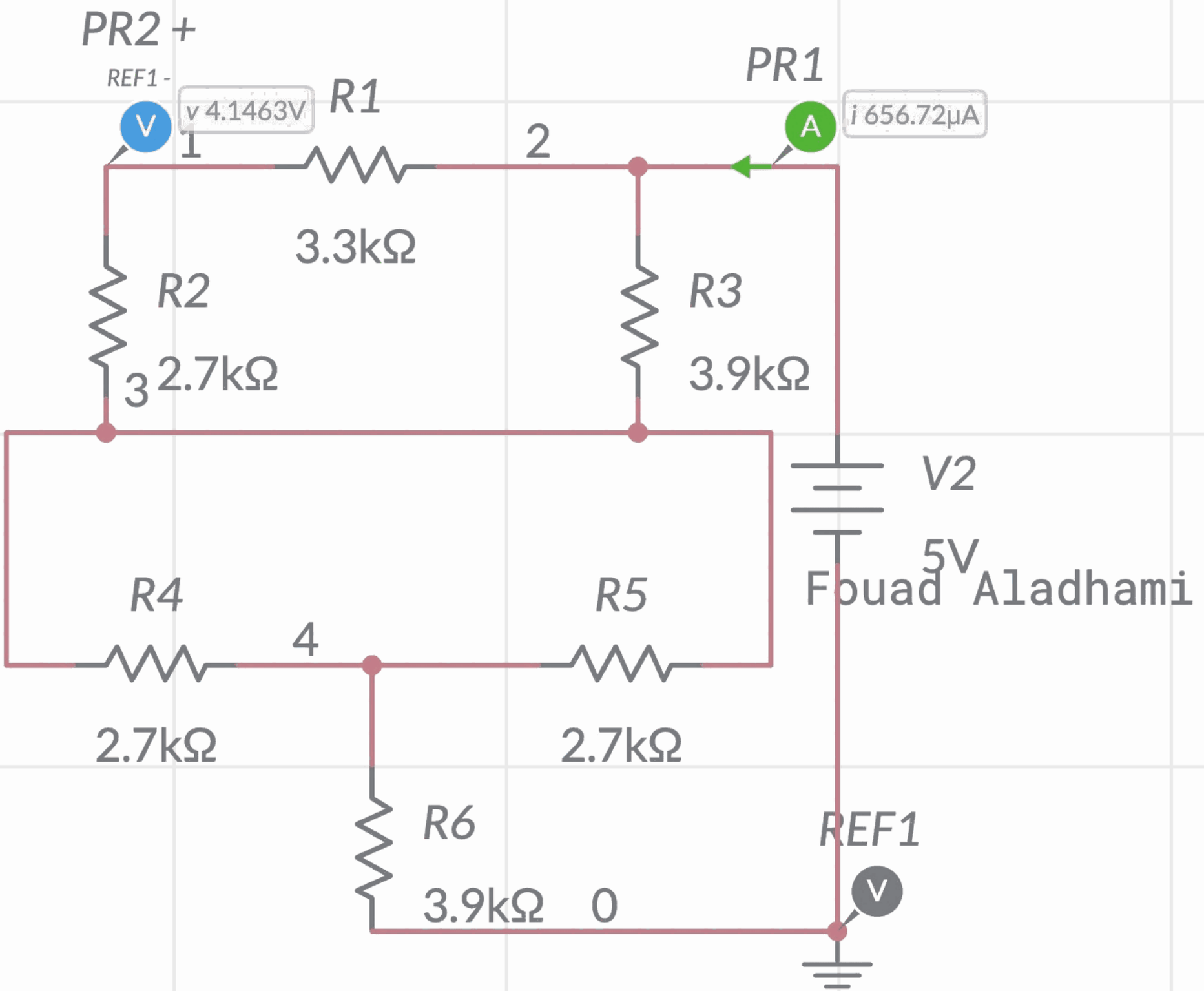
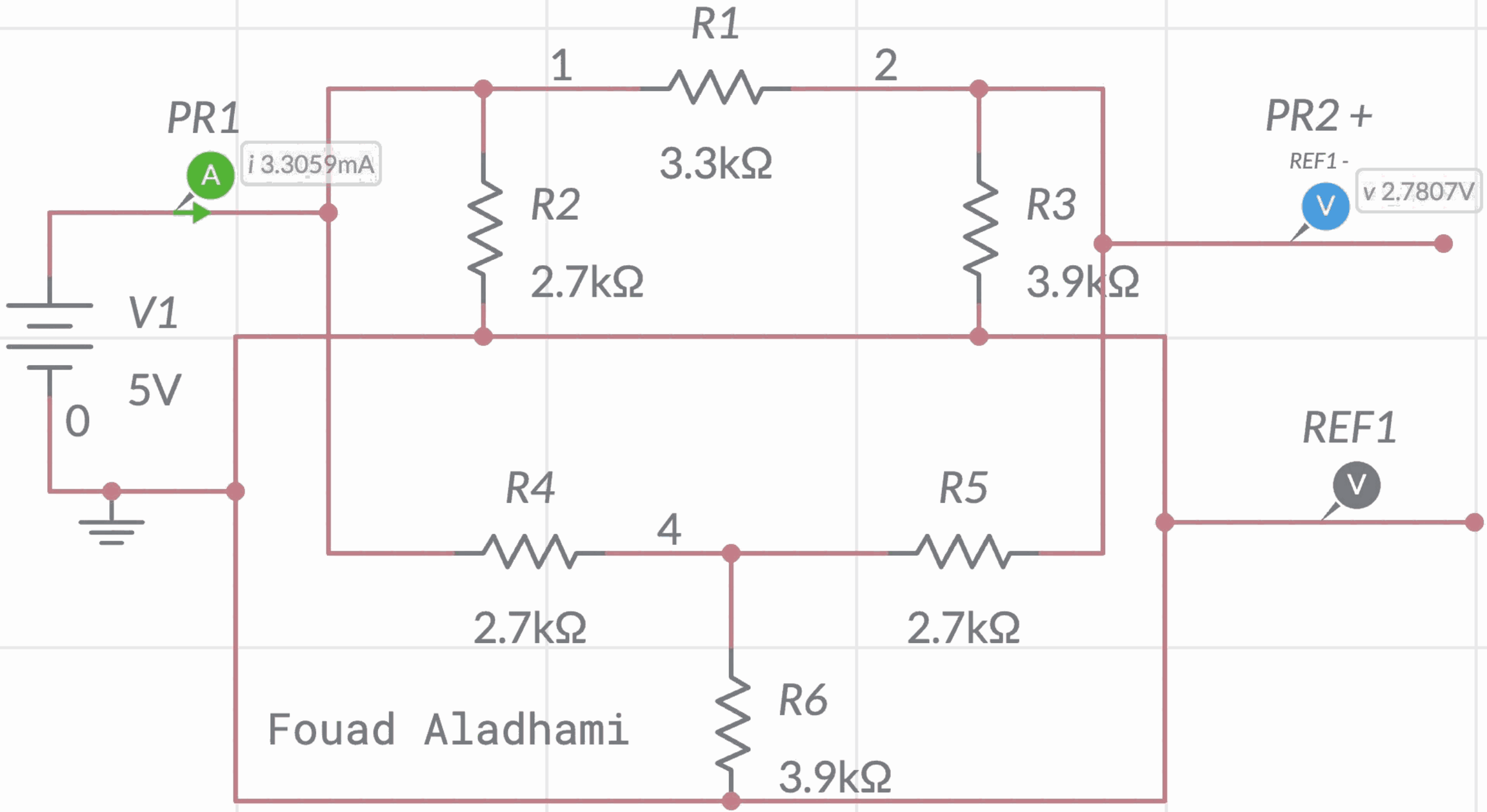
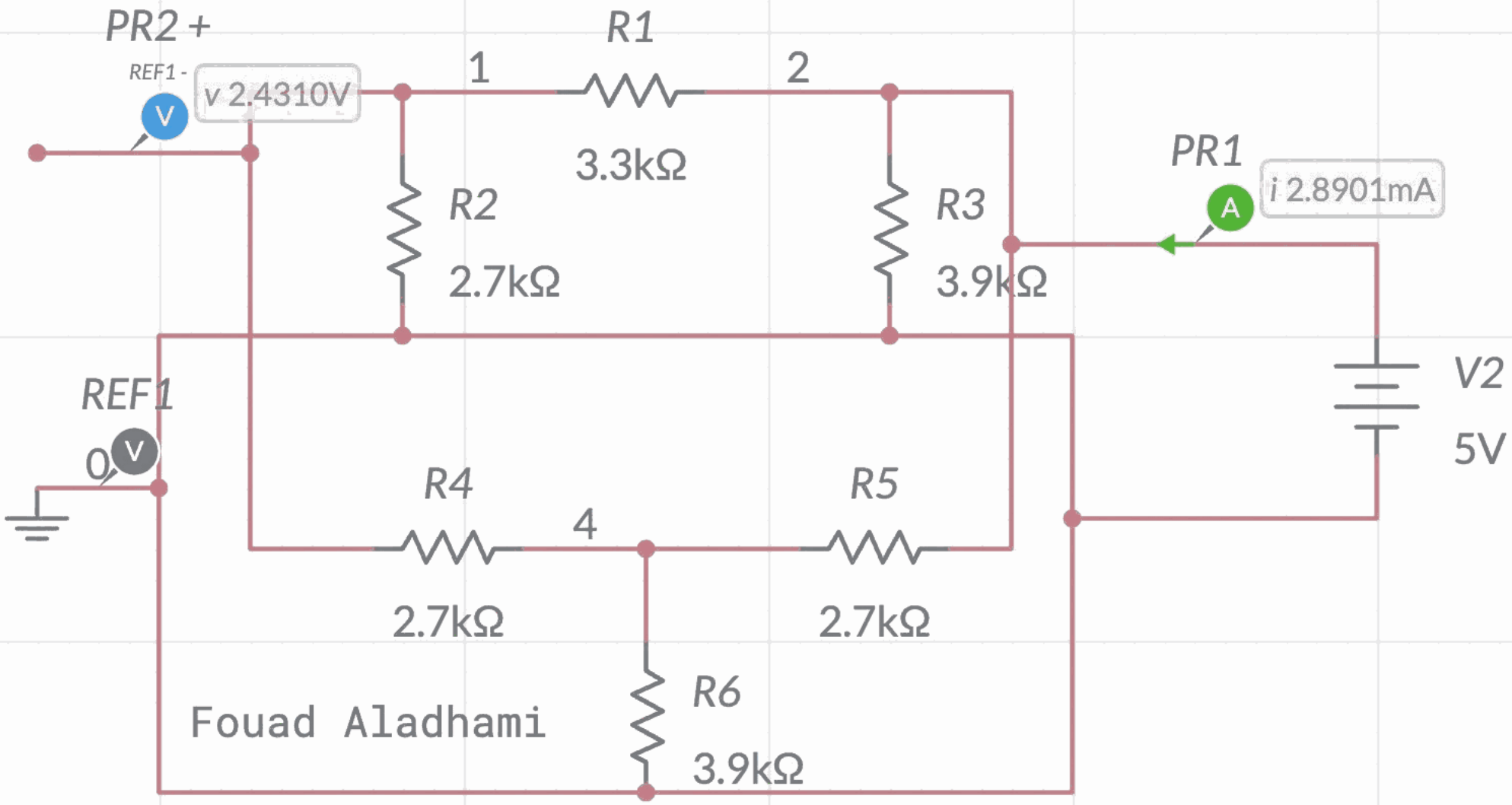
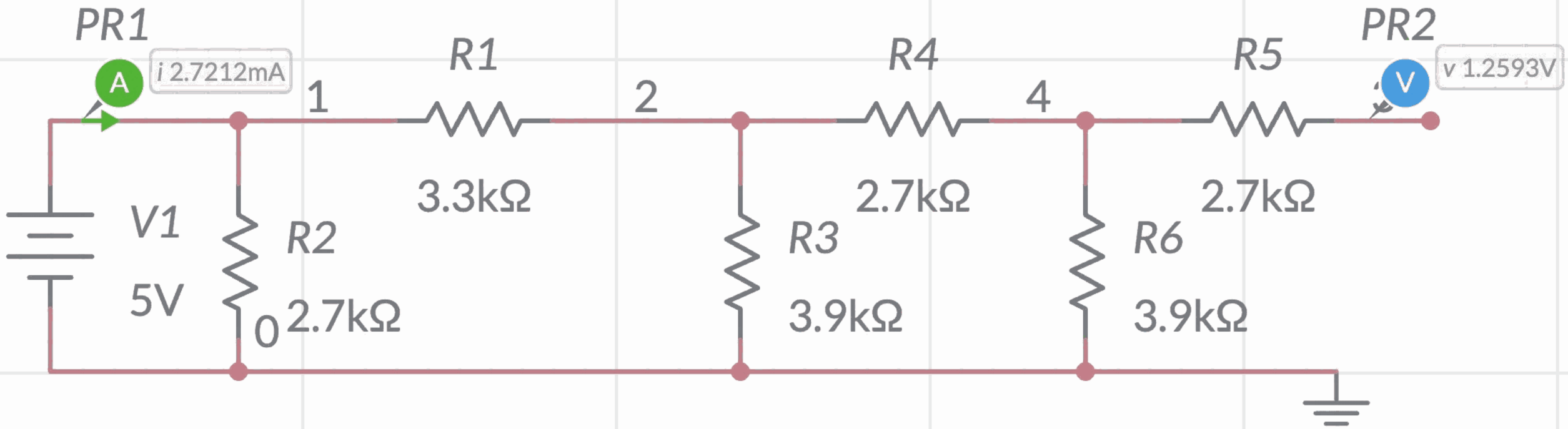

PR2 +
REF1 -
v 4.3762V
R1
1
2
3.3kΩ
PR1
i 693.13µA
R2
2.7kΩ
3
R3
3.9kΩ
V1
5V
R4
4
R5
Fouad Aladhami
2.7kΩ
2.7kΩ
R6
REF1
0
3.9kΩ

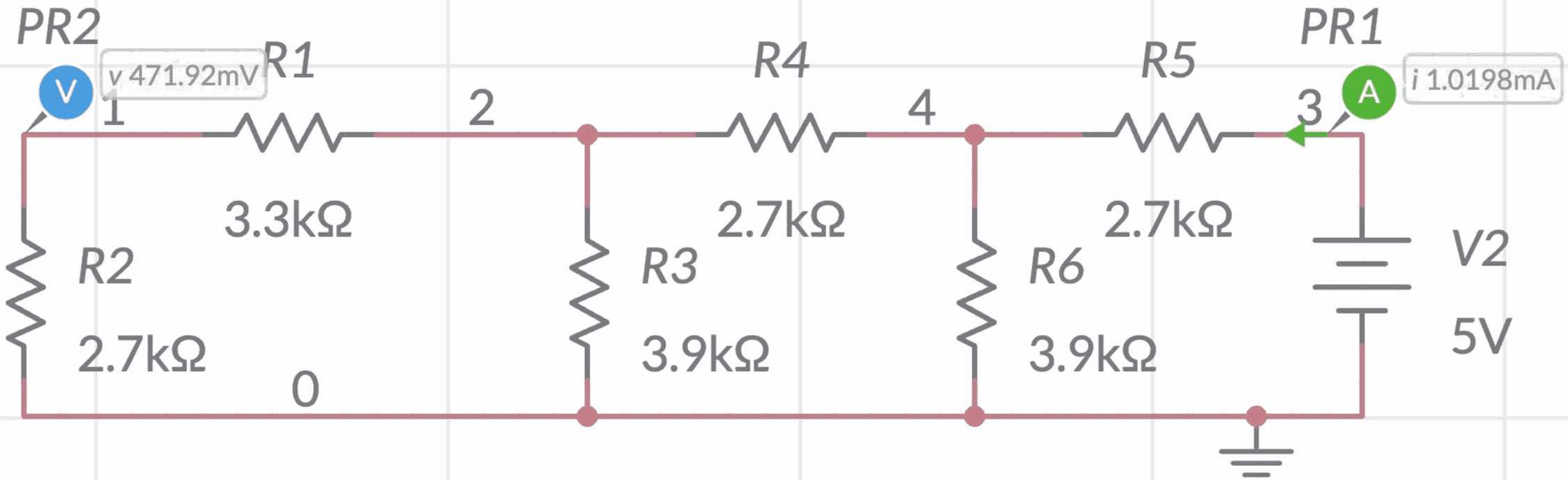

PR2 +
REF1 -
v 4.1463V
R1
3.3kΩ
1
2
PR1
i 656.72µA
R2
2.7kΩ
3
R3
3.9kΩ
V2
5V
Fouad Aladhami
R4
2.7kΩ
4
R5
2.7kΩ
R6
3.9kΩ
0
REF1


R1
1
2
3.3kΩ
PR1
A
i 3.3059mA
R2
2.7kΩ
R3
3.9kΩ
PR2 +
REF1 -
V
v 2.7807V
V1
5V
0
REF1
V
R4
4
R5
2.7kΩ
2.7kΩ
Fouad Aladhami
R6
3.9kΩ


PR2 +
REF1 -
v 2.4310V
R1
1
2
3.3kΩ
R2
2.7kΩ
R3
3.9kΩ
PR1
i 2.8901mA
V2
5V
REF1
0
R4
4
R5
2.7kΩ
2.7kΩ
Fouad Aladhami
R6
3.9kΩ



Fouad Aladhami



Fouad Aladhami

```
R1 = 3.3*10^3;
R2 = 2.7*10^3;
R3 = 3.9*10^3;
R13 = R1+ R3;
Req1 = (R2.*R13)/(R2+R13);
Z11 = Req1;
RI3 = R2./(R2+R13);
Z21 = R3*RI3;
%%
R12 = R1 + R2;
ZRI1 = R3./(R12+R3);
Z12 = ZRI1 .* R2;
Z22= R3.*(R12./(R12+R3));
%%

Z = [Z11, Z12;Z21,Z22];
Zdel = det(Z);
Y = [Z22./Zdel, -Z12./Zdel; -Z21./Zdel, Z11./Zdel ];
h = [Zdel./Z22, Z12./Z22; -Z21./Z22, 1./Z22 ];
g = [1./Z11,-Z12./Z11;Z21./Z11,Zdel./Z11];
T = [Z11./Z21, Zdel./Z21; 1./Z21, Z22./Z21];
%%
R4 = 2.7.*10^3;
R5 = 2.7.*10^3;
R6 = 3.9.*10^3;
Z11 = R4 + R6;
Z21 = R6;
Z12 = R6;
Z22 = R5 + R6;
Z = [Z11, Z12;Z21,Z22];
Zdel = det(Z);
Y = [Z22./Zdel, -Z12./Zdel; -Z21./Zdel, Z11./Zdel ];
h = [Zdel./Z22, Z12./Z22; -Z21./Z22, 1./Z22 ];
g = [1./Z11,-Z12./Z11;Z21./Z11,Zdel./Z11];
T = [Z11./Z21, Zdel./Z21; 1./Z21, Z22./Z21];
```

```
%%
Z11 = 5/(693.13*10^-6);
Z21 = 4.736/(693.13*10^-6);
Z12 = 4.1463/(656.72*10^-6);
Z22 = 5/(656.72*10^-6);
Z = [Z11, Z12;Z21,Z22];
Zdel = det(Z);
Y = [Z22./Zdel, -Z12./Zdel; -Z21./Zdel, Z11./Zdel ];
h = [Zdel./Z22, Z12./Z22; -Z21./Z22, 1./Z22 ];
g = [1./Z11,-Z12./Z11;Z21./Z11,Zdel./Z11];
T = [Z11./Z21, Zdel./Z21; 1./Z21, Z22./Z21];
%%
Z11 = 5/(3.3059*10^-3);
Z21 = 2.78/(3.3059*10^-3);
Z12 = 2.431/(2.8901*10^-3);
Z22 = 5/(2.8901*10^-3);
Z = [Z11, Z12;Z21,Z22];
Zdel = det(Z);
Y = [Z22./Zdel, -Z12./Zdel; -Z21./Zdel, Z11./Zdel ];
h = [Zdel./Z22, Z12./Z22; -Z21./Z22, 1./Z22 ];
g = [1./Z11,-Z12./Z11;Z21./Z11,Zdel./Z11];
T = [Z11./Z21, Zdel./Z21; 1./Z21, Z22./Z21];
%%
Z11 = 5/(2.721*10^-3);
Z21 = 2.78/(2.721*10^-3);
Z12 = 471.9/(1.0198);
Z22 = 5/(1.0198*10^-3);
Z = [Z11, Z12;Z21,Z22];
Zdel = det(Z);
Y = [Z22./Zdel, -Z12./Zdel; -Z21./Zdel, Z11./Zdel ];
h = [Zdel./Z22, Z12./Z22; -Z21./Z22, 1./Z22 ];
g = [1./Z11,-Z12./Z11;Z21./Z11,Zdel./Z11];
T = [Z11./Z21, Zdel./Z21; 1./Z21, Z22./Z21];


```